SUMIT KUMAR BHARTI

Data Science • Machine Learning • AI Engineering

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LinkedIn • GitHub • LeetCode • Credly • BeyondWord • NextGrade

PROFESSIONAL SUMMARY

Final-year Computer Science & Engineering (AI & Data Science) student who builds, evaluates and ships ML systems end to end — from data cleaning and feature engineering to model comparison, REST/FastAPI serving and deployed frontends. Hands-on across classical ML (scikit-learn), deep learning (PyTorch, TensorFlow/Keras), NLP and multilingual transformers, and GenAI (RAG, LLM integration, agents, LoRA/quantization, RLHF/SFT). Completed a Machine Learning Engineering internship at FlyRank AI, earned 171 Google Cloud skill badges (Diamond League), and finished Top 10 at a college hackathon. Seeking a Data Scientist / ML Engineer / AI Engineer role.

AT A GLANCE

CGPA	LeetCode	Contest Rating	GCP Badges	Projects	Hackathons
7.98 / 10	**341 solved**	**1,460**	**171** (Diamond League)	**6** built / deployed	**Top 10** — VIVEKA 2k26

EXPERIENCE

Machine Learning Engineering Intern — FlyRank AI (Remote) | *July 2026 – August 2026*

- Worked on ML models and real-world AI systems through structured assignment tracks, preparing and validating data before model training and evaluation.
- Completed hands-on modules on Vertex AI, LoRA/quantization, RLHF/SFT, GenAI agents and RAG workflows; all work version-controlled and submitted via GitHub (flyrank-ml-internship).
- Wrote and debugged Python for data ingestion, cleaning and transformation steps feeding model training and evaluation.

PROJECTS

1. Emotion Reader (BeyondWord) — Multilingual AI Emotion Recognition | Live • Source

- Built an emotion and sentiment recognition system that classifies text into six core emotions and scores psychometric VAD dimensions (Valence, Arousal, Dominance), with empathy and communication guidance for each reading.
- Hybrid transformer model in PyTorch served through a FastAPI backend; supports 100+ languages including English, Hindi, Hinglish, Spanish and French, with calibrated probability outputs.
- Single-text and batch CSV analysis modes, session history and copyable response templates; React 19 + TanStack frontend deployed on Vercel.

Tech Stack: *Python, PyTorch, Transformers, FastAPI, React 19, TanStack, Vercel*

2. GyanVerse AI — RAG-Based Intelligent QnA System | Final-Year Project (In Progress)

- Building a Retrieval-Augmented Generation pipeline for semantic search and question answering over document data using Sentence Transformers and FAISS/ChromaDB, with GPT-4o-mini for answer generation.
- Streamlit interface for real-time, context-aware responses; developed with a 3-member team under faculty guidance using Git.

Tech Stack: *Python, Sentence Transformers, FAISS, ChromaDB, GPT-4o-mini, Streamlit, Git*

3. RailSaarthi — Train ETA Prediction & Control-Room Platform | Smart India Hackathon 2026 (Team Garuda X, SIH26028)

- Designed a multi-factor ETA prediction engine, delay root-cause classifier, connecting-train miss-risk analyser and control-room zonal dashboard for Indian Railways.
- Owned data handling and model evaluation for the ML components within a 6-member team.

Tech Stack: *Python, scikit-learn, TypeScript, TanStack Start*

4. Student Performance Prediction — ML Web App & REST API | nextgrade.netlify.app

- Predicts academic performance and flags at-risk students early from study hours, attendance, prior scores, sleep and lifestyle factors on 1,200 student records.
- Compared Linear Regression, Random Forest and Gradient Boosting; selected Gradient Boosting (R^2 0.85, 5-fold CV R^2 0.83) after hyperparameter tuning.
- Served real-time predictions via a Flask REST API with a StandardScaler pipeline; interactive Chart.js radar-chart frontend.

Tech Stack: *Python, scikit-learn, Flask, JavaScript, Chart.js, Netlify*

5. Airline Delay Prediction — ML Classification | GitHub

- End-to-end pipeline on a 124K-row imbalanced flight dataset (34.4% delayed): ingestion, cleaning, validation, EDA and feature engineering (e.g. Departure_Hour).

- Benchmarked Logistic Regression, Decision Tree and Random Forest using F1-score and ROC-AUC to handle class imbalance; selected Random Forest (ROC-AUC 0.67).

Tech Stack: *Python, Pandas, NumPy, scikit-learn, Matplotlib*

6. KarigarSetu — The Artisan's Ledger | VIVEKA: The Intelligence 5.0, Techfest 2k26 — **Top 10 Finish** (Team CodeVengers)

- Platform connecting Chikankari artisans of Lucknow directly with buyers, with artisan/buyer modules, transparent pricing and authenticity badges. Built with AI-assisted tooling.

Tech Stack: *React, TypeScript, Tailwind CSS, Supabase, Vite*

Model Evaluation Summary

Project	Task	Models Compared	Selected	Key Metric
Student Performance	Regression	Linear Regression, Random Forest, Gradient Boosting	Gradient Boosting	R^2 0.85 • CV R^2 0.83
Airline Delay	Imbalanced binary classification	Logistic Regression, Decision Tree, Random Forest	Random Forest	ROC-AUC 0.67 • F1
Emotion Reader	Multi-class emotion + VAD	Hybrid transformer	Hybrid transformer	6 emotions • 100+ languages
GyanVerse AI	Retrieval + generation	FAISS / ChromaDB retrieval	RAG + GPT-4o-mini	In progress

RESEARCH

- Co-authored **"AI-Based Exam Proctoring System"** research paper, supervised by Mrs. Anamika Sharma (SRMU) — explored ML/computer-vision techniques for automated academic integrity monitoring.

TECHNICAL SKILLS

Area	Skills	Proficiency
Programming	Python	Advanced
	Java • SQL / MySQL	Intermediate
	JavaScript • TypeScript • React	Beginner
Machine Learning	Supervised & Unsupervised Learning • Feature Engineering • Model Evaluation (F1, ROC-AUC, R^2, cross-validation) • Hyperparameter Tuning	Strong
Deep Learning & NLP	PyTorch • TensorFlow / Keras • Transformers • Sentence Transformers • Multilingual NLP • Text Mining	Intermediate
GenAI & LLMs	RAG • FAISS • ChromaDB • GPT / LLaMA integration • Prompt Engineering • GenAI Agents • LoRA / Quantization • RLHF / SFT	Intermediate
Data Analysis & Viz	Pandas • NumPy • Matplotlib • Seaborn • EDA • Statistical Analysis • Power BI • Tableau	Strong
Cloud & MLOps	Google Cloud (Vertex AI, BigQuery ML, Cloud Run, Cloud Functions, IAM, Terraform, Binary Authorization) • Azure ML Studio • IBM Cloud / Watson Studio	Intermediate
Serving & Web	FastAPI • Flask • REST APIs • Streamlit • Chart.js	Intermediate
Tools	Git • GitHub • Jupyter / Colab • VS Code • MS Excel • Google Sheets	Strong

COMPETITIVE PROGRAMMING

Platform	Solved	Easy	Medium	Hard	Contest Rating
LeetCode	**341**	94	184	63	1,460

Languages: Python, Java, MySQL. Earned LeetCode 50-Day and 100-Day 2026 badges.

EDUCATION

Qualification	Institution	Year	Score
B.Tech — CSE (AI & Data Science)	Shri Ramswaroop Memorial University (SRMU), Barabanki, UP	2023 – 2027	CGPA 7.98 / 10 (SGPA 8.26)
Senior Secondary (CBSE, PCM)	PM Shri Kendriya Vidyalaya, Lucknow, UP	2023	75.4%
Secondary (CBSE)	Army School, Binaguri, West Bengal	2021	87.4%

Relevant Coursework: Machine Learning, Data Science, Statistical Analysis, Artificial Intelligence, Big Data Analytics, Data Visualisation, Data Structures & Algorithms, DBMS, OOP, Operating Systems.

CERTIFICATIONS

Domain	Certification	Issuer	Score / Date
Data Science	Google Data Analytics Professional Certificate	Coursera / Google	94.87%
Data Science	Google Advanced Data Analytics Capstone	Coursera / Google	100%
Data Science	Foundations of Data Science	Coursera / Google	99% • Jul 2026
Data Science	The Power of Statistics	Coursera / Google	97.91% • Jul 2026
Data Science	Go Beyond the Numbers: Translate Data into Insights	Coursera / Google	100%
Data Science	Foundations: Data, Data, Everywhere	Coursera / Google	Jul 2026
Data Science	Data Analytics with Python (Elite)	NPTEL, IIT Roorkee	65% • Jan–Apr 2026
Data Science	Data Analysis with Python (DA0101EN) • Data Science Methodology (DS0103EN)	IBM Skills Network	Oct 2026
ML / AI	Machine Learning Pipelines with Azure ML Studio	Coursera	97.22% • Jul 2026
ML / AI	Foundations of AI and Machine Learning	Microsoft / Coursera	100%
ML / AI	Predictive Modeling Fundamentals I (PA0101EN) • Deep Learning Fundamentals (ML0115EN)	IBM Skills Network	Oct 2026
ML / AI	Artificial Intelligence Analyst (CEAIA1IN) • Learning Path: Data Science & Artificial Intelligence	IBM Skills Network / SRMU	Oct 2026
ML / AI	NLP and Text Mining	Simplilearn SkillUp	Apr 2026
GenAI	Generative AI and AI Agent learning plan (LLMs, Prompt Engineering, AI Agents, RAG)	IBM SkillsBuild	Jul 2026
GenAI	Generative AI Essentials — AI for All	MPIT-CoE & TCS Foundation via TCS iON	Apr 2026
GenAI	AI Fluency: Framework & Foundations • Introduction to Claude • Claude 101	Anthropic	—
GenAI	GEAR: Intro to Agents • Scaling Agents Across the Enterprise • Building Production-Ready Agents • Gemini Enterprise Agent Ready	Google	Badges
Cloud	Google Cloud Skills Boost — 171 skill badges, Diamond League	Google Cloud	2026
Simulations	Quantium Data Analytics • Tata GenAI-Powered Data Analytics • Deloitte Australia Data Analytics	Forage	Job simulations

HACKATHONS & ACHIEVEMENTS

Event	Role / Project	Result
VIVEKA: The Intelligence 5.0, Techfest 2k26 (SRMU)	Team CodeVengers — KarigarSetu	**Top 10**
Smart India Hackathon 2026 — SRMU Internal Round	Team Garuda X — RailSaarthi (SIH26028, Ministry of Railways)	Internal round
Smart India Hackathon 2025 — SRMU Internal Round	Team NextGen Coders — Smart Bus Tracking app	Internal round

LEADERSHIP & ACTIVITIES

- **Joint Secretary & Management Head — Social Media Club, SRMU** (Oct 2025 – Feb 2026): tracked engagement metrics across 10+ university events to guide outreach; coordinated cross-functional teams for large-scale student events (200+ attendees).
- **Volunteer — Internal Hackathon, SRMU** (Feb – Mar 2025): managed participant coordination and team logistics for a 100+ student hackathon.
- **Communities:** Google Developer Group (GDG) Lucknow member; Google Developer Program premium member.
- **Workshops & Training:** Aptitude & DSA Training Program (SRDT, Jan 2026) • Jumpstart ML: Python Made Easy (IEEE Student Chapter & Tech Fusion Club, Oct 2024) • Role of AI in Achieving the SDGs (SRMU, Nov 2024) • Net Zero 2070 (SRMU, Mar 2025).